

CY 541: Fundamentals of Organic Chemistry

Course Code	CY 541
Course Name	Fundamentals of Organic Chemistry
L-T-P-C	3-0-0-3
Prerequisites	B.Sc. (with Chemistry) or Teachers consent
Intended for	UG/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Structure and Bonding**

Review of basic principles of structure and bonding, application of acid base concepts, HSAB theory, aromaticity and antiaromaticity, Hckels rule, anti-aromaticity, y-aromaticity, homo- aromaticity n-annulenes, heteroannulene, fullerenes, C-60, cryptates, Bonds weaker than covalent; addition compounds, inclusion compounds, crown ethers, cyclodextrins, catenanes and rotaxanes. [10 Lectures]

- **Stereochemistry**

Conformational analysis of alkanes and cycloalkanes, effect of conformation on reactivity. Elements of symmetry, chirality, molecules with more than one chiral center, projection formulae (i) Fischer (ii) Sawhorse (iii) Newman (iv) Flying Wedge; tLectureseo and erytLectureseo isomers, methods of resolution, optical purity, enantiotopic and diastereotopic atoms, groups and faces, stereospecific and stereoselective synthesis, Asymmetric synthesis. Optical activity in the absence of chiral carbon. Axis and planar chirality. Absolute configuration of chiral centers. [12 Lectures]

- **Reaction mechanism**

Structure and Reactivity: Types of mechanisms, types of reactions, thermodynamic and kinetic requirements, Hammond postulate, Curtin-Hammett principle, transition states and intermediates, methods of determining mechanisms, isotopic effects. Generation, structure, stability and reactivity of carbocations, carbanions, free radicals, carbenes and nitrenes. Effect of structure on reactivity. The Hammett equation and linear free energy relationship (sigma-rho) relationship, Taftequation. [12 Lectures]

- **Oxidation, Reduction and Rearrangements** [8 Lectures]

Oxidation: Different oxidative processes. Hydrocarbons, carbonyl compounds, amines, hydrazines and sulphides.

Reduction: Different reductive processes. Hydrocarbons, carbonyl compounds, nitro, nitroso, azo and oxime groups. Hydrogenolysis.

Rearrangements: General mechanistic considerations, nature of migration, migratory aptitude, nucleophilic, electrophilic and free radical rearrangement. A detailed study of various arrangements reactions.

Text Books

1. F. A. Carey and R. A. Sundberg, **Advanced Organic Chemistry, Part A: Structure and Mechanisms**, 5th edition, Springer, 2007.
2. D. Nasipuri, **Stereochemistry of Organic Compounds**, Wiley, 1994.
3. E. J. Eliel, **Stereochemistry of Carbon Compounds**, McGraw Hill

References

1. D. G. Morris, **Stereochemistry**, RSC Tutorial Chemistry Text 1, 2001.
2. S. H. Pine, **Organic Chemistry**, McGraw Hill, 1987.
3. J. March, **Advanced Organic Chemistry**, John Wiley & Sons, 1992.